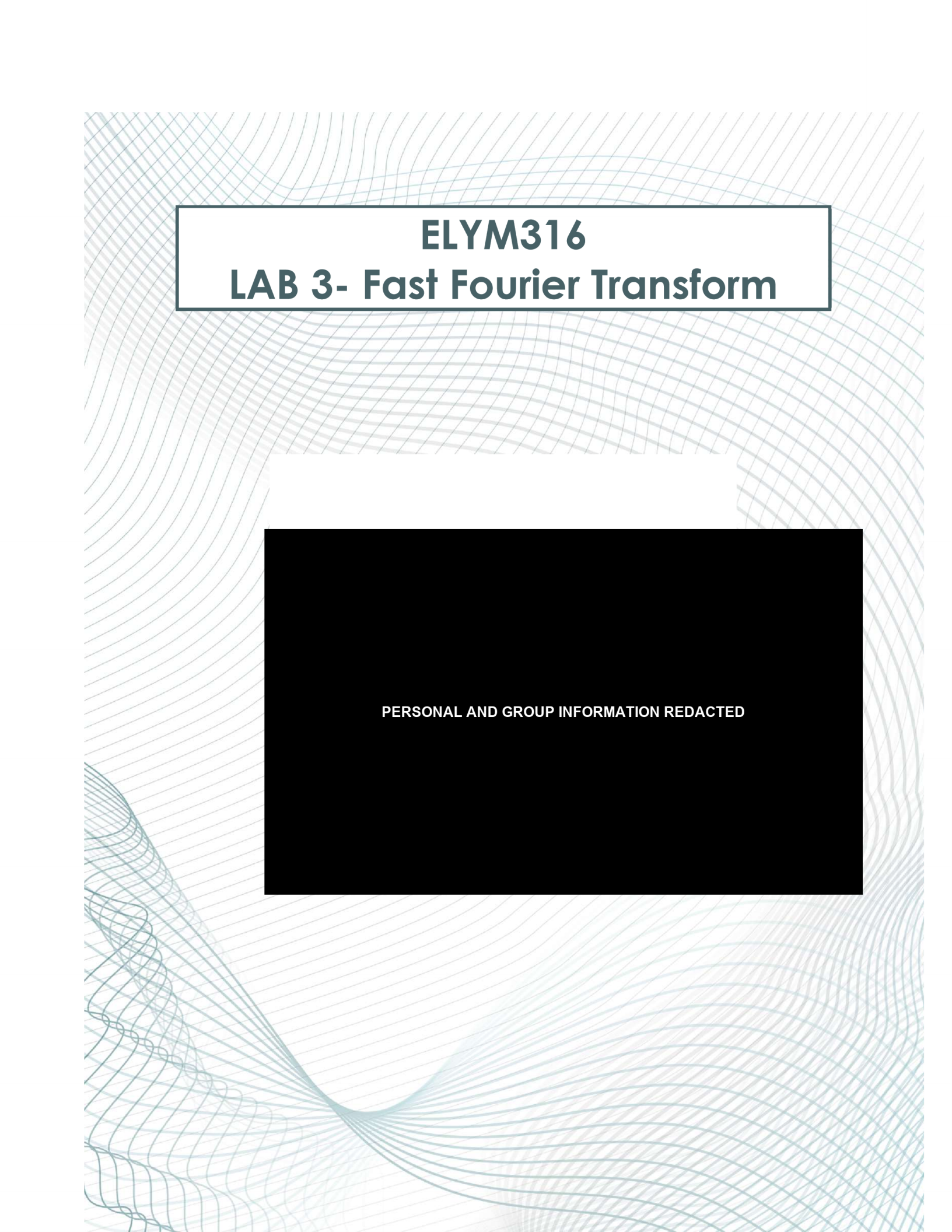
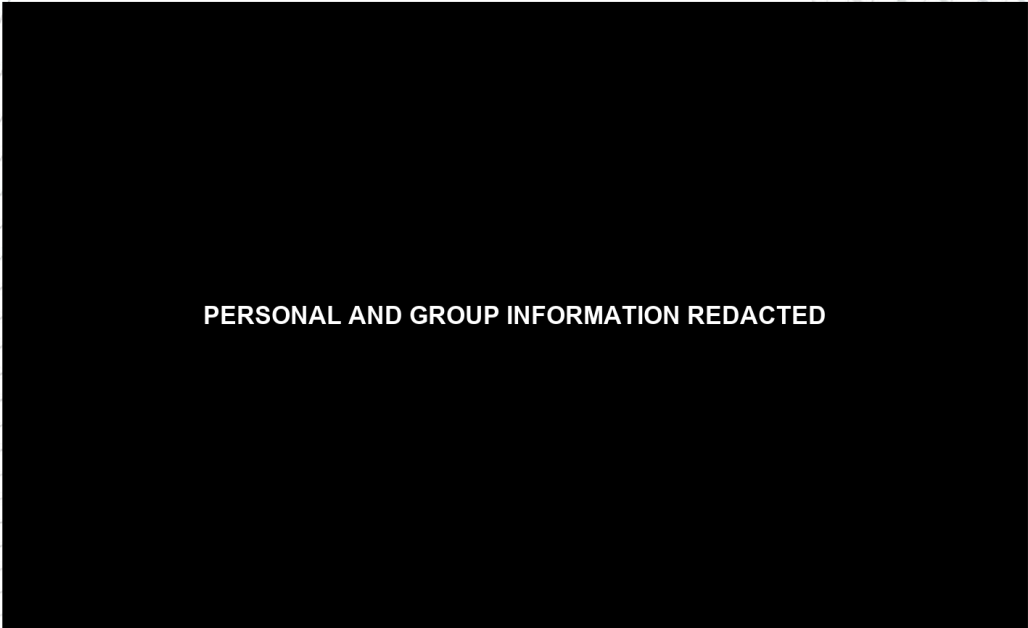




ELYM316

LAB 3- Fast Fourier Transform



PERSONAL AND GROUP INFORMATION REDACTED

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Part 1 - Individual signals (1 second, fs=1000 Hz for clean plots)

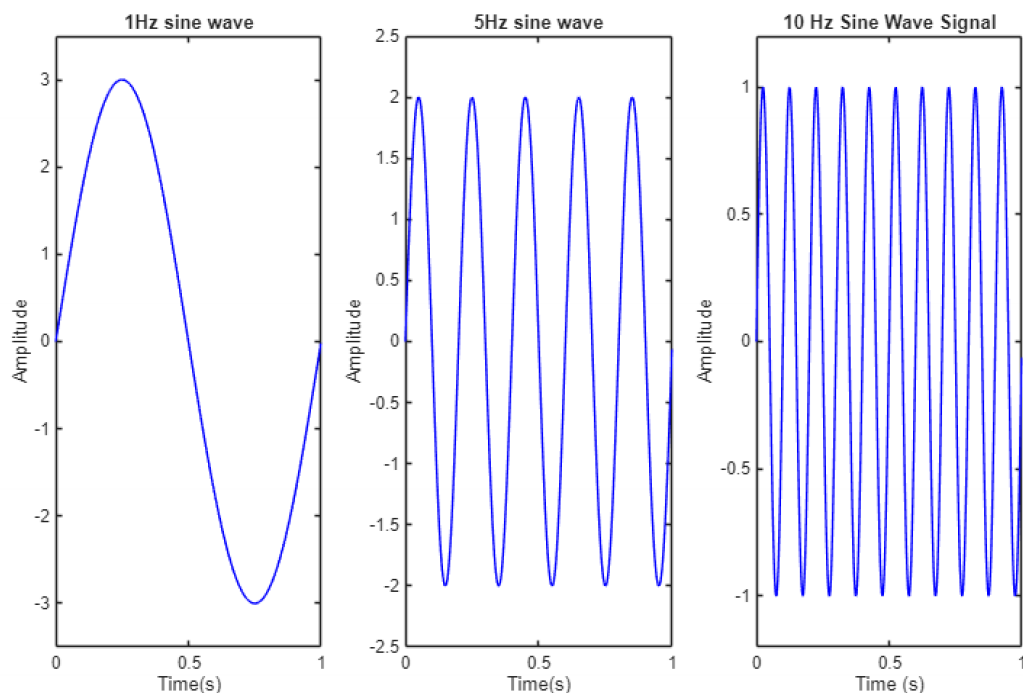
```
fs1 = 1000; t1 = 0:1/fs1:1-1/fs1;

s1 = 3*sin(2*pi*1*t1);    % 1 Hz, amplitude 3
s2 = 2*sin(2*pi*5*t1);    % 5 Hz, amplitude 2
s3 = 1*sin(2*pi*10*t1);   % 10 Hz, amplitude 1

figure;
subplot(1,3,1); plot(t1,s1,'b'); title('1Hz sine wave');
xlabel('Time(s)'); ylabel('Amplitude'); ylim([-3.5 3.5]);

subplot(1,3,2); plot(t1,s2,'b'); title('5Hz sine wave');
xlabel('Time(s)'); ylabel('Amplitude'); ylim([-2.5 2.5]);

subplot(1,3,3); plot(t1,s3,'b'); title('10 Hz Sine Wave Signal');
xlabel('Time (s)'); ylabel('Amplitude'); ylim([-1.2 1.2]);
```

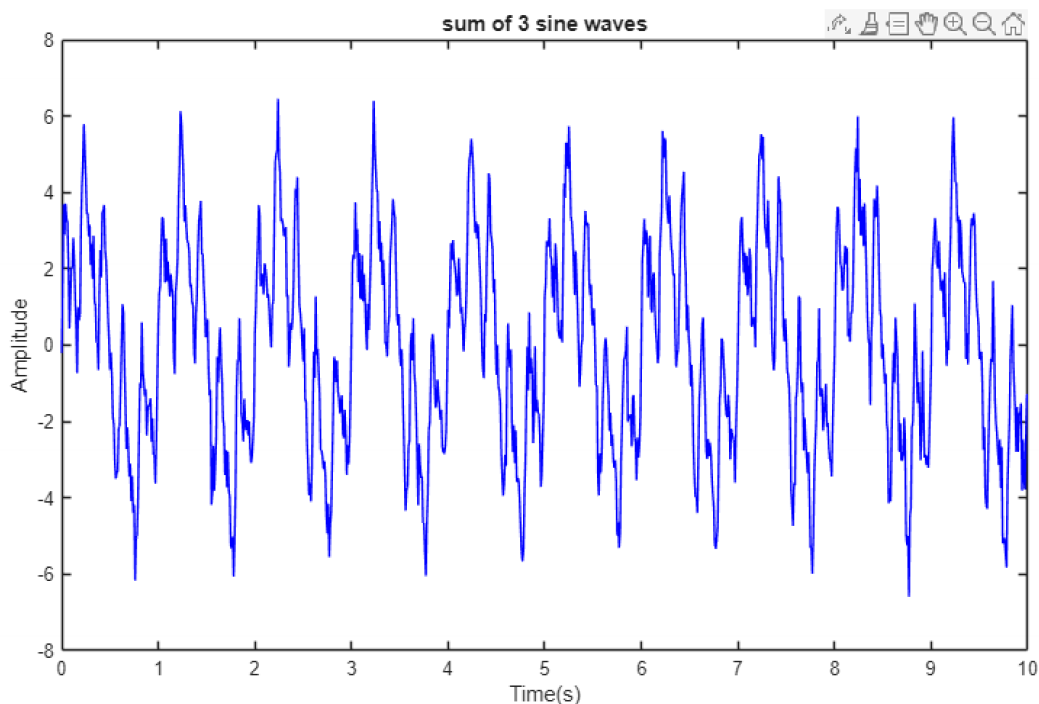


Part 2 - Composite + noise (10 seconds, fs=100 Hz)

```
fs = 100; t = 0:1/fs:10-1/fs; % 1000 samples total

composite = 3*sin(2*pi*1*t) + 2*sin(2*pi*5*t) + 1*sin(2*pi*10*t);
noisy = composite + 0.5*randn(size(t)); % add Gaussian noise

figure;
plot(t, noisy, 'b'); title('sum of 3 sine waves');
xlabel('Time(s)'); ylabel('Amplitude');
```

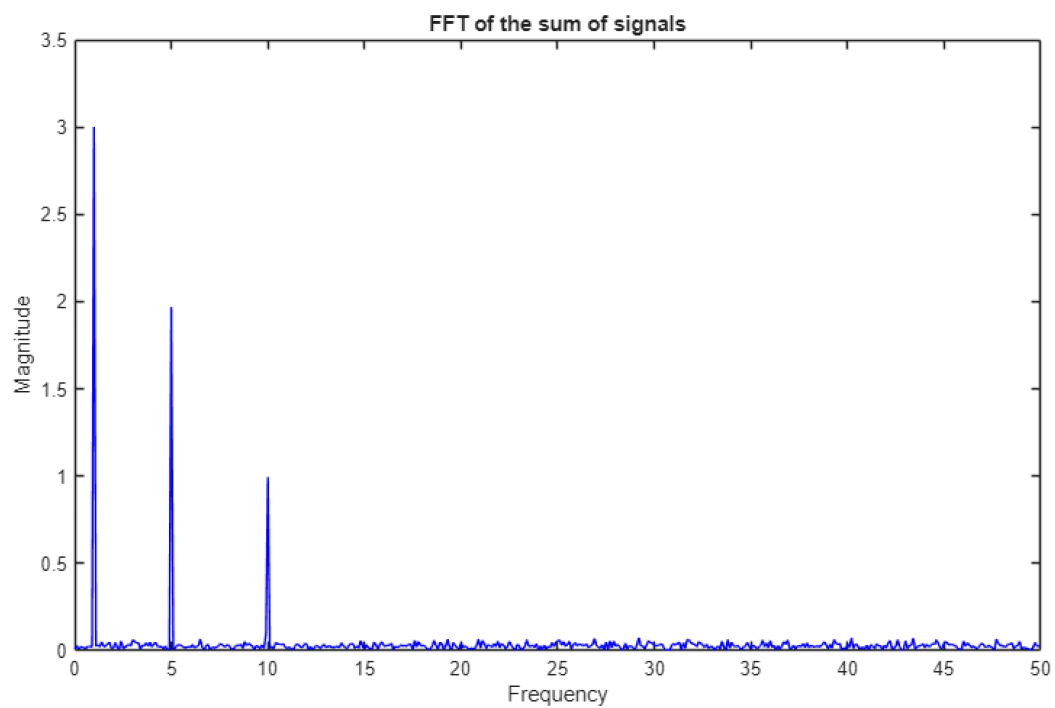
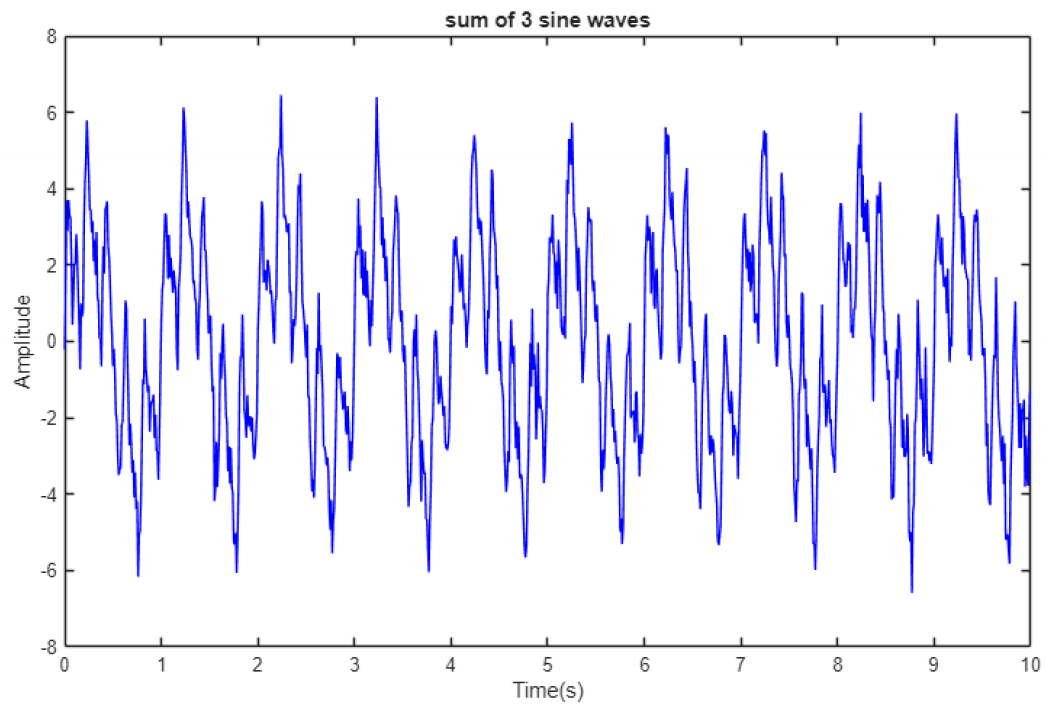


Part 3 - FFT

```
N = length(noisy);
Y = fft(noisy);
f = (0:N-1)*(fs/N); % frequency axis

magnitude = (2/N)*abs(Y); % scale to recover true amplitudes
magnitude(1) = magnitude(1)/2; % DC component not doubled

figure;
plot(f(1:N/2), magnitude(1:N/2), 'b');
title('FFT of the sum of signals');
xlabel('Frequency'); ylabel('Magnitude');
xlim([0 50]);
```



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